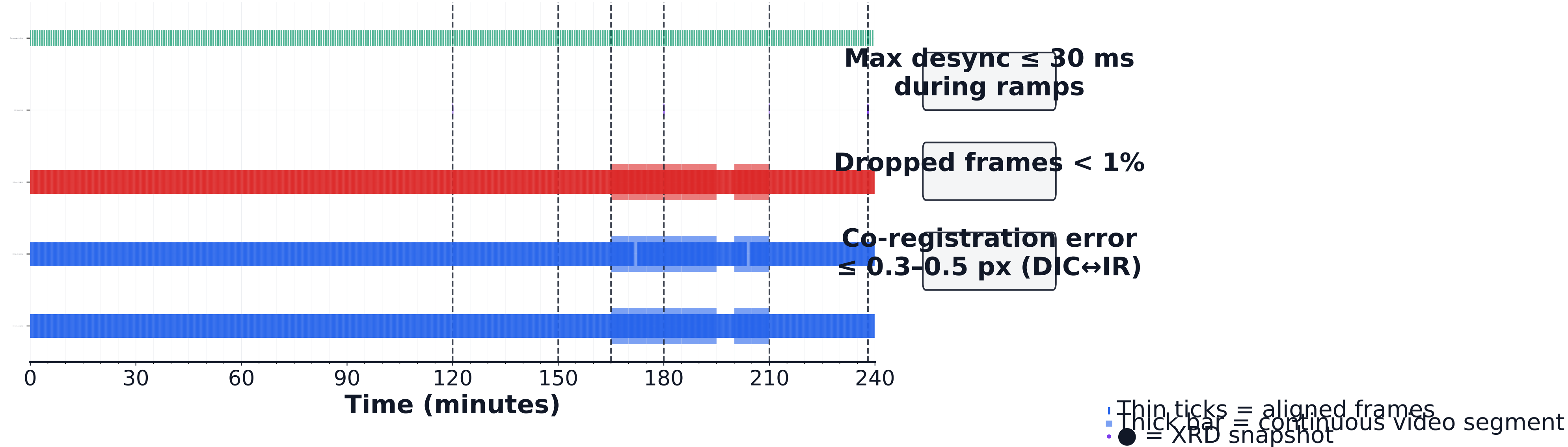


Synchronized Time Raster — DIC, IR, XRD (Aligned Frames & Continuous Capture)

Unified clock; hardware trigger; Δt alignment

XRD snapshotPre-holdStart rampLoadPeak loadPost-shock



Accessibility: Horizontal timeline from 0 to 240 minutes with five stacked rows for DIC A, DIC B, IR, XRD, and thermocouples. Thin ticks align across rows at the same times. Thick bars on DIC and IR indicate continuous recording from 165–195 and 200–210 minutes. XRD shows discrete dots at 120, 180, 210, and 238 minutes. Vertical dashed lines label key events. Side notes report ≤ 30 ms desynchronization and $< 1\%$ dropped frames.